

Lab Experiment: 3

Title: Data Mining and Data Warehousing Using Tools for DMQL and OLAP

Course title: Data Mining Problem Description

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Submitted to-

Md. Humayun Kabir

Professor

Department of Computer Science and Engineering
Jahangirnagar University

&

Dr. Md Musfique Anwar

Professor

Department of Computer Science and Engineering
Jahangirnagar University
Savar, Dhaka-1342

Class Roll	Exam Roll	Name
383	210903	Abdullah Nazmus-Sakib
379	210899	Shah Sultan
391	210911	Md Khaled Mahmud
395	210915	Tawhidul Islam

Project Title:

Intelligent Thesis Supervisor Recommendation and Performance Prediction System (ThesisHub)

1. Introduction

Modern academic institutions generate large volumes of data related to student performance, thesis supervision, and evaluation activities. Traditional databases are optimized for transaction processing but are not suitable for analytical processing and knowledge discovery.

Data Mining and Data Warehousing techniques help transform raw academic data into meaningful insights. In this experiment, Data Mining Query Language (DMQL) and OLAP tools are used to analyze ThesisHub data to support supervisor recommendation and student performance prediction.

2. Objective

The main objectives of this experiment are:

- To apply Data Mining techniques using DMQL queries
 - To perform multidimensional data analysis using OLAP operations
 - To extract useful academic patterns from ThesisHub dataset
 - To support decision making for thesis supervision and evaluation
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3. Tools Used

- Data Warehouse (Star Schema Dataset)
 - SQL / DMQL Query Tools
 - OLAP Analysis Tool (Power BI / Excel Pivot / Tableau)
 - Python (Optional for Mining Validation)
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4. Dataset Used

Dataset includes:

Student Data

- Student_ID
- CGPA
- Department
- Attendance_Percentage

Teacher Data

- Teacher_ID
- Expertise_Area
- Evaluation_Rating

Thesis Performance Data

- Final_Marks
 - Submission_Duration
 - Feedback_Score
 - Task_Completion_Percentage
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5. Data Warehousing Concept Used

Star Schema Model

Fact Table:

- Thesis_Performance_Fact

Dimension Tables:

- Student_Dimension
 - Teacher_Dimension
 - Thesis_Dimension
 - Time_Dimension
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6. DMQL (Data Mining Query Language) Implementation

DMQL is used to extract patterns and relationships from warehouse data.

DMQL Query 1 — High Performance Students

```
SELECT Student_ID, CGPA, Final_Marks
FROM Combined_Dataset
WHERE CGPA > 3.5 AND Final_Marks > 80;
```

Purpose:

Identify students likely to perform well in thesis work.

DMQL Query 2 — Best Performing Supervisors

```
SELECT Teacher_ID, AVG(Final_Marks)
FROM Combined_Dataset
GROUP BY Teacher_ID;
```

Purpose:

Identify supervisors producing best thesis results.

DMQL Query 3 — At-Risk Students

```
SELECT Student_ID, CGPA, Attendance_Percentage
FROM Combined_Dataset
WHERE CGPA < 3.0 OR Attendance_Percentage < 75;
```

Purpose:

Detect students needing academic support.

7. OLAP Operations

OLAP allows multidimensional data analysis.

7.1 Roll-Up (Aggregation)

Example:

Average Marks by Department

```
SELECT Department, AVG(Final_Marks)
FROM Combined_Dataset
GROUP BY Department;
```

7.2 Drill-Down (Detailed Analysis)

Example:

Marks by Teacher → Then by Student

7.3 Slice Operation

Example:

Analyze only AI Domain Theses

```
SELECT *
FROM Combined_Dataset
WHERE Research_Domain = 'AI';
```

7.4 Dice Operation

Example:

Filter by CGPA Range + Domain:

```
SELECT *
FROM Combined_Dataset
WHERE CGPA BETWEEN 3.0 AND 4.0
AND Research_Domain = 'Data Science';
```

8. Results and Findings

From DMQL and OLAP analysis:

- Students with CGPA > 3.5 perform significantly better
 - Teachers with higher evaluation ratings produce better results
 - Late submission strongly correlates with low final marks
 - High attendance improves thesis completion success
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9. Applications in ThesisHub

Supervisor Recommendation

Uses:

- Teacher performance history
- Student academic profile

Performance Prediction

Uses:

- CGPA
 - Attendance
 - Submission duration
 - Task completion rate
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10. Advantages of DMQL and OLAP

- Fast data retrieval
 - Multidimensional analysis
 - Pattern discovery
 - Decision support
 - Easy visualization
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11. Conclusion

This experiment successfully demonstrated the use of Data Mining and Data Warehousing tools for analyzing academic data. DMQL queries helped extract performance patterns, while OLAP operations enabled multidimensional academic analysis. The results support intelligent supervisor recommendation and thesis performance prediction in the ThesisHub system.